

Neuro-Glove Rehabilitation Suite

Clinical Games and Exercise Protocol Guide (Glove-Guided Telemetry)

PROTOCOL VERSION v2.4 - Approved	HARDWARE PROFILE StretchSense Wireless Glove	INPUT MODALITY Pure Sensor Telemetry (100%)	CLINICAL TARGET Upper Limb Neuro-Motor Recovery
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1. Executive Clinical Protocol Matrix

The Neuro-Glove Rehabilitation Suite translates therapeutic upper-extremity motor rehabilitation into interactive, biofeedback-driven 3D environments. Each game directly maps to an isolated or compound anatomical motion protocol. All patient actions are executed strictly through natural hand, finger, and wrist articulations without keyboard or mouse requirements.

Game Title	Rehabilitation Exercise	Primary Biomechanical Motions	Target Clinical Pathologies
Gelato Tower	Forearm Pronation / Supination	Internal & external rotation around forearm axis (neutral 0° to ±45°)	Post-stroke spasticity, distal radius fracture recovery, pronator/supinator contracture
Balloon Blitz	Hand Open / Close (Grasp & Extension)	MCP & IP flexion (functional grasp) and full digital extension (open hand)	Radial nerve palsy, extensor weakness, grasp-release deficit, flexor hypertonia
Pizza Spin	Wrist Circumduction	Continuous 360° circular carpal rotation (flexion, deviation, extension)	Carpal tunnel post-op, radiocarpal contracture, limited functional arc of motion
Rolling Wonder	Compound Multi-Motion Agility	Lateral wrist tilt steering, rapid extension jump, 360° circumduction turbo	Complex motor re-learning, cognitive-motor dual tasking, dynamic stabilization

CORE CLINICAL PRINCIPLES ACROSS THE SUITE

- Quantitative Range of Motion Measurement:** Real-time capture of degree angles, bend percentages, and circular trajectory completion rates directly from glove hardware.
- Inhibitory Control & Spasticity Mitigation:** Targeted thresholds (such as balloon burst limits and cone tilt limits) discourage uncontrolled spastic bursts in favor of graded, voluntary muscle recruitment.
- Automated Telemetry Logging:** Complete repetition counts, session durations, and accuracy metrics are stored to track longitudinal patient recovery.
- Ergonomic Isolation:** Designed for supported arm positioning to prevent synergistic shoulder abduction or trunk compensation tricks.

2. Game 1: Gelato Tower — Forearm Pronation / Supination

Forearm rotation is fundamental for daily tasks including feeding, key turning, and utensil manipulation. Post-stroke flexor synergy frequently confines patients to fixed pronation. Gelato Tower challenges patients to achieve active, controlled rotational ROM to catch falling gelato scoops on an upright cone.



Figure 1: Gelato Tower live session displaying tilt gauge HUD and clinical pronation/supination badges.

TARGETED MUSCLE RECRUITMENT

- **Pronation:** Pronator teres, pronator quadratus.
- **Supination:** Supinator, biceps brachii.
- **Neutral Recenter:** Brachioradialis.

SENSOR TELEMETRY MECHANISM

- **Detection:** Inertial Measurement Unit (IMU) roll angle.
- **Dynamic Range:** -45° (pronation) to +45° (supination).
- **Neutral Baseline:** 0° vertical thumb-up posture.

HOW TO PLAY WITH REHABILITATION GLOVES (GLOVE-ONLY CONTROLS)

1. **Ergonomic Starting Posture:** Forearm rested on therapy table with elbow flexed to 90°. Thumb points vertically toward the ceiling (0° baseline).
2. **Forearm Pronation** **PRONATION** : Internally rotate the forearm so the palm turns downward toward the table. The cone tilts smoothly to the left to catch leftward falling scoops.
3. **Forearm Supination** **SUPINATION** : Externally rotate the forearm so the palm turns upward toward the ceiling. The cone tilts smoothly to the right to catch rightward falling scoops.
4. **Dynamic Balancing:** Return to 0° neutral to center the cone. Counter-balance dynamic tower sway as scoops accumulate.

Progression (Levels 1 to 5): Level 1 (Vanilla Coast, 8 scoops) • Level 2 (Pistachio Park, 12 scoops) • Level 3 (Berry Mountain, 16 scoops) • Level 4 (Golden Sunset, 20 scoops) • Level 5 (Grand Gelateria, 25 scoops at rapid velocity).

3. Game 2: Balloon Blitz — Hand Open / Close

Functional hand rehabilitation requires coordination between finger flexion (cylindrical grasp) and active digital extension (releasing objects). In patients with radial nerve injury or post-CVA extensor paresis, voluntary release is significantly delayed. Balloon Blitz enforces alternating flexion pumping and complete digital extension releases.

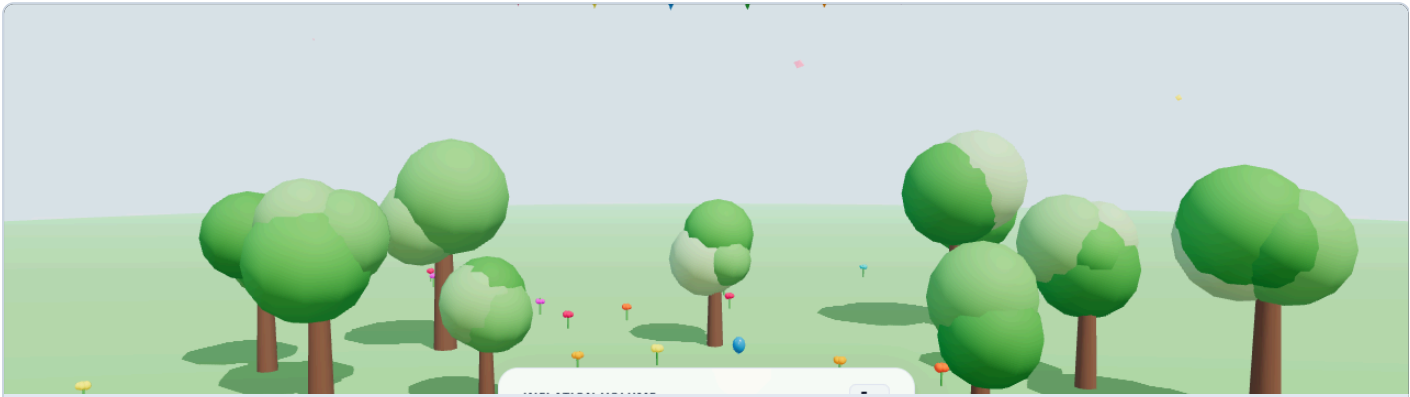


Figure 2: Balloon Blitz live session displaying real-time inflation volume meter and target green window.

TARGETED MUSCLE RECRUITMENT

- **Flexion (Grasp):** Flexor digitorum superficialis, profundus, lumbricals.
- **Extension (Open Hand):** Extensor digitorum communis, extensor indicis.

SENSOR TELEMETRY MECHANISM

- **Detection:** 5-channel stretch bend sensors.
- **Grasp Threshold:** >55% multi-digit joint bend.
- **Release Threshold:** <25% joint bend (full extension).

HOW TO PLAY WITH REHABILITATION GLOVES (GLOVE-ONLY CONTROLS)

1. **Ergonomic Starting Posture:** Forearm supported on armrest or table in neutral posture; fingers relaxed in semi-flexed resting state.
2. **Hand Grasp (Pump Air) GRASP :** Close all four fingers and thumb firmly into a functional fist. Each grasp actuates the pneumatic pump piston, increasing balloon volume. Rhythmic pumping allows gradual accumulation.
3. **Controlled Target Window TARGET :** Pump balloon until the telemetry bar enters the **Target Green Window (70% to 85%)**. Pumping past 95% triggers burst limit, training voluntary inhibitory control.
4. **Hand Extension (Launch Balloon) EXTEND :** Spread all fingers and thumb wide into maximal extension. The balloon is sealed and launches into the sky.

Progression (Levels 1 to 5): Level 1 (Meadow Carnival, 6 balloons, wide window) • Level 2 (Spring Fair, 9 balloons) • Level 3 (Skyway Fiesta, 12 balloons) • Level 4 (Twilight Bazaar, 15 balloons) • Level 5 (Grand Gala, 18 balloons, tight 75%-83% window).

4. Game 3: Pizza Spin — Wrist Circumduction

Wrist circumduction represents a compound motion combining flexion, extension, radial deviation, and ulnar deviation. Following wrist fracture or carpal tunnel release, wrist trajectories often become angular or guarded. Pizza Spin trains continuous, smooth circular coordination around the radiocarpal pivot.



Figure 3: Pizza Spin live session showing 360° circular telemetry gauge and artisan pizza table.

TARGETED MUSCLE RECRUITMENT

- **Flexors:** Flexor carpi radialis, flexor carpi ulnaris.
- **Extensors:** Extensor carpi radialis longus/brevis, carpi ulnaris.
- **Coordinated Transition:** Smooth sequential motor firing through 360°.

SENSOR TELEMETRY MECHANISM

- **Detection:** Multi-axial carpal angular integration.
- **Repetition Unit:** 1 full 360° circular trajectory = 1 repetition.
- **Direction:** Supports clockwise and counter-clockwise modes.

HOW TO PLAY WITH REHABILITATION GLOVES (GLOVE-ONLY CONTROLS)

1. **Ergonomic Starting Posture:** Forearm supported horizontally on therapy table; wrist positioned freely past table edge with palm facing down.
2. **Tracing Smooth Circles** **CIRCUMDUCTION** : Without translating the elbow or shoulder, guide the hand through a circular pathway: extension → radial deviation → flexion → ulnar deviation → back to extension.
3. **Repetition Tracking** **360° ROTATION** : The circular gauge fills from 0% to 100%. Completing 360° registers 1 repetition.
4. **Stretching Crust to Target** **TARGET** : Each repetition spins and stretches the pizza dough toward target diameter (14 inches), delivering gourmet ingredients.

Progression (Levels 1 to 5): Level 1 (Trattoria Rustica, 12", 4 rotations) • Level 2 (Napoli Classic, 14", 6 rotations) • Level 3 (Firenze Gourmet, 16", 8 rotations) • Level 4 (Venezia Master, 18", 10 rotations) • Level 5 (Grand Pizzeria, 20", 12 rotations).

5. Game 4: Rolling Wonder — Compound Agility

Rolling Wonder bridges isolated exercises into rapid compound agility. Pip the hamster travels inside a transparent exercise sphere along a colorful track requiring continuous lane steering, obstacle jumping, and rapid speed bursts.



Figure 4: Rolling Wonder live track runner displaying multi-lane agility and glove motion status.

TARGETED MUSCLE RECRUITMENT

- **Steering:** Carpal deviators (radial and ulnar deviation).
- **Leaping:** Digital extensors (extensor digitorum communis).
- **Turbo Whirl:** Multi-axial carpal rotators in rapid burst.

SENSOR TELEMETRY MECHANISM

- **Sensor Fusion:** IMU deviation angle + digital flex channels.
- **Active State HUD:** ROLLING CRUISE, EXTENSION JUMP, TURBO WHIRL.
- **Speed Dynamic:** Base 30 km/h to 45 km/h turbo velocity.

HOW TO PLAY WITH REHABILITATION GLOVES (GLOVE-ONLY CONTROLS)

- **Lateral Wrist Tilt (Steering)** **WRIST TILT** : Tilt wrist left or right (radial/ulnar deviation or pronation/supination bias) to maneuver between lanes, collecting gems and dodging hurdles.
- **Rapid Hand Extension (Jump)** **EXTEND** : Snap fingers and thumb wide open from rest to execute a vertical leap over road blocks and mushrooms.
- **Circular Wrist Whirl (Turbo)** **360° CIRCLE** : Rotate wrist in a rapid circumduction cycle to trigger Rainbow Turbo boost, accelerating forward velocity to 45 km/h.

Progression (Levels 1 to 5): Level 1 (Sunny Meadow, 120m, 800 pts) • Level 2 (Daisy Fields, 150m, 1,200 pts) • Level 3 (Crystal Stream, 180m, 1,600 pts) • Level 4 (Twilight Hills, 210m, 2,000 pts) • Level 5 (Rainbow Peak, 250m, 2,500 pts).

6. Clinical Best Practices & Session Guidelines

To ensure optimal therapeutic outcomes and prevent compensatory movement patterns during sensor-guided glove exercises, supervising therapists should adhere to the following clinical standards:

- ### 1. PRE-SESSION TARE & CALIBRATION

Before initiating any game module, instruct the patient to assume their natural, resting neutral posture with fingers semi-flexed and thumb upward. Verify through the glove connectivity status modal that all 5 sensor bend channels read within resting baseline parameters and the IMU roll angle is centered at 0°.
- ### 2. GUARDING AGAINST SYNERGISTIC COMPENSATION

Post-stroke hemiparetic patients commonly compensate for restricted forearm supination by abducting the shoulder or leaning the torso laterally. Ensure the elbow remains stabilized at 90 degrees against the torso throughout Gelato Tower and Pizza Spin. Utilize an elbow support block if unprompted shoulder elevation occurs.
- ### 3. MONITORING SPASTIC FATIGUE & CLENCHING

Observe the patient for signs of flexor synergy fatigue (such as involuntary fist clenching, delayed release in Balloon Blitz, or accidental curling in Rolling Wonder). Schedule brief 30-to-60 second rest pauses between level completions to prevent hypertonicity or cramping.
- ### 4. LONGITUDINAL TELEMETRY DOCUMENTATION

All repetitions, completed rotational degrees, grasp-extension cycles, and session durations are recorded automatically to the clinical directory. Clinicians can review progress trend charts across longitudinal therapy sessions to adjust prescribed difficulty parameters.

PRESCRIBED WEEKLY PROTOCOL RECOMMENDATION			
Day	Primary Focus	Prescribed Game	Target Dosage
Monday	Forearm Rotational ROM	Gelato Tower (Levels 1-3)	3 sets of 15 scoops caught
Tuesday	Grasp & Active Extension	Balloon Blitz (Levels 1-3)	3 sets of 10 launched balloons
Wednesday	Carpal Circumduction	Pizza Spin (Levels 1-2)	4 sets of 8 completed 360° circles
Thursday	Compound Coordination	Rolling Wonder (Levels 1-2)	3 runs of 150m course
Friday	Combined Evaluation	Full Suite Rotation	1 set per protocol module